Convex Perturbation and the Suppression of the Indivisibility Deficit

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Abstract

We study how the Chapman–Kolmogorov (CK) deficit of a column-stochastic two-time family $\{\Gamma(t \leftarrow s)\}_{s \le t}$ responds to convex mixing with a divisible reference family: $\Gamma_g = (1-g)\,\Gamma_0 + g\,\Gamma_{\mathrm{ref}}$, $g \in [0,1]$. The reference models a dynamically consistent, state-conditioned interaction with the process; its divisibility is the structural hypothesis of the construction. The organizing result is an exact algebraic identity for the deficit operator of the mixed family, $\Delta(t,s,r;\Gamma_g) = (1-g)(A-CE) - g(1-g)(C-D)(F-E)$—where A, C, E denote the substrate kernels on the time pairs (t,s), (t,r), (r,s) and D, F the corresponding reference kernels—valid for every finite column-stochastic pair, which exhibits the perturbed deficit as a residual substrate term plus a substrate–reference interaction term and yields the operator-norm suppression bound $D(t,s,r;\Gamma_g) \le (1-g)\,D(t,s,r;\Gamma_0) + g(1-g)\,M(t,s,r)$. The response need not be monotone in g—the norm bound can rise before it falls when the interaction term dominates, and the exact deficit can vanish and rebound in the interior of the coupling range—and this nonmonotonicity forces a two-threshold formulation of detectability: a first-crossing threshold $g_{\mathrm{first}}(\delta t;\varepsilon)$ and a terminal suppression threshold $g_{\mathrm{term}}(\delta t;\varepsilon)$, the coupling beyond which the deficit at temporal resolution δt remains below the detection threshold ε at every stronger coupling. We prove the explicit universal bound $g_{\mathrm{term}}(\delta t;\varepsilon) \le 1 - \varepsilon/[D_{\mathrm{CK}}(\delta t;\Gamma_0) + M_*(\delta t)]$, so that arbitrarily strong suppression is reached at a computable sub-unit coupling, and $0 < g_{\mathrm{first}} \le g_{\mathrm{term}}$. Integrated over the trajectory simplex on $L^2([0,T]\times\Omega)$, the deficit functional obeys $J(\Gamma_g) = O((1-g)^2)$. On any compact interval of resolutions on which the unperturbed deficit remains uniformly above ε, the first-crossing threshold admits an explicit strictly positive lower bound, uniform in the resolution—an interval-local stability result for the detectability of indivisibility under weak coupling, with the product $g_{\mathrm{first}} \cdot \delta t$ bounded below accordingly. We state plainly that this is a compactness-based stability statement and not a dynamical speed limit. The analysis operates on $L^2([0,T]\times\Omega)$ with one-sided causal operators, and no time-reversal symmetry is assumed. A worked two-state example exhibits the substrate-dominant and interaction-dominant regimes, including a nonmonotone rebound, in closed form. We refer to g as the observer coupling; the name is a label inherited from the stochastic–quantum correspondence and carries no interpretive weight in any statement or proof.

Keywords. Indivisible stochastic processes; Chapman–Kolmogorov; non-Markovian dynamics; convex perturbation; deficit suppression; detection threshold; one-sided causal operators.

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1 Introduction

A two-time family $\{\Gamma(t \leftarrow s)\}_{s \le t}$ of column-stochastic matrices on a finite configuration space $\Omega = \{1, \dots, N\}$ encodes empirical transition statistics of an Ω-valued process via $\Gamma_{ij}(t \leftarrow s) = \mathbb{P}(X_t = i \mid X_s = j)$. The family satisfies the Chapman–Kolmogorov (CK) factorization condition if

$$\Gamma(t \leftarrow s) \;=\; \Gamma(t \leftarrow r)\,\Gamma(r \leftarrow s) \qquad \text{for all } 0 \le s < r < t \le T. \tag{1}$$

A family that fails (1) on a set of triples of positive measure is *indivisible* in the sense of [5, 3, 4], and the operator-theoretic structure of indivisibility—the singular geometry of the composition-deficit operator, the local indivisibility tensor, the Stinespring dilation of the empirical kernel, and the causal organization on $L^2([0,T] \times \Omega)$—has been developed in the companion work [1]. An independent foundational argument for indivisibility as a structural feature of physical state descriptions is given in [2].

The present paper concerns a question that the spectral framework of [1] does not address: how does the empirical detectability of the CK deficit depend on a perturbation of the family that models an interaction with the process? The interactions modeled are those that are both conditioned on the instantaneous state at the observation time *and* dynamically consistent across time—a measurement protocol, a control input, a state-feedback rule whose action composes coherently along intermediate times. Such an interaction is modeled by a *divisible* reference family; the divisibility is the structural hypothesis of the model, made explicit in Definition 2.5, and is not a consequence of state conditioning alone, since a collection of state-conditioned kernels need not compose across time without the consistency requirement. Mixing the indivisible substrate with a divisible reference in convex combination yields a perturbed family whose CK deficit is controlled by that of the substrate weighted by the residual mass. The observation is elementary at the level of definitions; its quantitative consequences are not, and one of them—the failure of the response to be monotone in the coupling—reorganizes the detectability theory around two distinct thresholds.

The principal contributions are four. First, we formalize the convex perturbation as the canonical model of a dynamically consistent state-conditioned interaction with an indivisible family (Section 2, Definition 2.6). The choice of a convex one-parameter perturbation rather than a more elaborate operator-valued perturbation is structural: it captures the essential mechanism—a divisible component enters in measure g at the expense of the indivisible substrate in measure $1 - g$—without coupling parameters that would obscure the operator-norm response of the deficit. Second, we establish the exact algebraic identity for the deficit operator of the perturbed family (Theorem 3.1), which is the organizing result of the paper: it factors the perturbed deficit into a residual substrate term of first order in $1 - g$ and a substrate–reference interaction term of order $g(1 - g)$, and it makes the possible nonmonotonicity of the response an exact algebraic fact rather than an artifact of estimation (Remark 3.2). Third, the nonmonotonicity forces a two-threshold formulation of empirical detectability: the first-crossing threshold g_{first}, at which the deficit first dips below a detection threshold ε, and the terminal suppression threshold g_{term}, beyond which it remains below ε for every larger coupling. The terminal threshold is the operationally meaningful quantity, and the main theorem (Theorem 4.3) establishes the explicit universal bound $g_{\text{term}} \le 1 - \varepsilon/[D_{\text{CK}} + M_*]$ together with $0 < g_{\text{first}} \le g_{\text{term}}$; the integrated trajectory-space deficit is suppressed quadratically, $J(\Gamma_g) = O((1-g)^2)$ (Theorem 5.3). Fourth, we prove an interval-local stability result (Proposition 4.9): on any compact interval of resolutions on which the unperturbed deficit remains uniformly detectable, the first coupling at which detectability is lost is bounded below by an explicit positive constant, so the product $g_{\text{first}} \cdot \delta t$ is uniformly bounded below there. We interpret this result honestly: it is a compactness-based stability statement about weak coupling, not a resolution–coupling scaling law; the comparison with product-form bounds in dynamical speed-limit theory is confined to

a single paragraph of the discussion (Section 7.3), where its limitations are stated.

The paper is structured as follows. Section 2 establishes the framework: column-stochastic two-time families, the deficit operator and its norms, and the divisible reference. Section 3 states and proves the deficit identity and the suppression bound. Section 4 defines the two thresholds, proves the terminal suppression theorem and the interval-local stability result, and works a two-state example through both regimes. Section 5 lifts the construction to one-sided causal operators on $L^2([0,T]\times\Omega)$ and proves the quadratic suppression of the integrated deficit. Section 6 treats state-dependent coupling—now with a rigorous finite-dimensional bound—and nonlinear coupling. Section 7 positions the results against the related literature.

Causal structure. The construction throughout operates on $L^2([0,T]\times\Omega)$ with one-sided causal operators: propagation of densities is time-forward by construction, and no symmetry under time reversal is assumed or required. The retarded transition operators of the substrate and reference families ([1, Definition 7.1]) are causal, and the perturbed retarded operator $\mathcal{R}_g$ of Definition 5.1 inherits this causality (Proposition 5.2). We make this explicit at the outset to forestall a reading of the Hilbert-space methods used here as implying time-reversal symmetry where none is intended.

Relationship to prior work. The mathematical frame of this paper sits at the intersection of three traditions.

Divisibility-based non-Markovianity in open dynamics. The non-Markovianity literature [10, 11, 12, 13, 14, 15] quantifies departure from divisibility through measures defined on quantum dynamical maps; the closest classical-side bridge is the joint quantum/classical analysis of Vacchini–Smirne–Laine–Piilo–Breuer [16], with the recent Benatti–Chruściński–Nichele [17] statement of how P-divisibility transfers between the commutative and non-commutative settings. The present paper operates upstream of these measures: rather than defining a non-Markovianity functional, we study how a structurally given deficit on a two-time column-stochastic family responds to a one-parameter perturbation. The connection is that the threshold behavior established here would induce, via any of the standard divisibility-based functionals, a perturbation-dependent measurability of non-Markovianity; that connection is recorded in Remark 7.1 but not pursued.

Convex perturbations of dynamical maps. The behavior of CK violation and divisibility under convex combinations of dynamical maps has been studied in the CPTP setting by Wudarski–Chruściński [18], Jagadish–Srikanth–Petruccione [19], Siudzińska [20], and Pathak–Shaji [21], with a resource-theoretic formulation by Bhattacharya–Bhattacharya–Majumdar [22]. The closest precedent for the present work is [22], which establishes free operations and a robustness measure of non-Markovianity under convex combinations on bounded time intervals. The contributions here differ on three counts. (i) The setting is a continuous two-time column-stochastic family, not a discrete-time CPTP convex combination. (ii) The perturbation mixes an indivisible substrate with a divisible reference rather than two CPTP families, and the algebraic structure that emerges is asymmetric: the interaction term in the deficit identity factors as a product of two substrate–reference differences, reflecting the divisibility of one summand. (iii) The threshold theory is formulated on the deficit at a fixed temporal resolution δt, not on a resource-theoretic functional defined on a fixed time interval.

Process-tensor formalism. The multi-time process-tensor framework of [23, 5, 24, 25] treats non-Markovian quantum dynamics through a Stinespring/dilation construction over discrete sequences of CPTP control operations. The present setting differs structurally on four counts. (i) The object of study is a continuous two-time family on a finite configuration space, not a discrete-time multi-instrument tensor. (ii) The kernels are column-stochastic, not arbitrary

CPTP. (iii) The perturbation is a one-parameter convex mixing rather than an inserted control operation. (iv) The classical-readout case (diagonal-CPTP, in the language of the diagonal-retention lift, [1, Definition 5.1]) is the exclusive scope here. The two frameworks are compatible: the present perturbation theorem specializes, in the discrete-time multi-instrument setting, to a statement about CPTP perturbations of process tensors, but the joint (t, s)-continuous formulation here makes no use of the process-tensor machinery and operates with column-stochastic objects throughout.

A structural comparison with product-form inequalities in classical and quantum speed-limit theory is deliberately confined to Section 7.3.

2 Framework

We work throughout with the framework of two-time column-stochastic transition families and the indivisibility deficit, following the conventions of [1], whose operator/norm notation we adopt exactly: Δ denotes the deficit *operator* and D its operator norm. Two results of that work are used as cited inputs and taken as given: the regularity conditions (R1)–(R2) of [1, Theorem 5.4 (Stinespring lift)], which are the boundary and continuity conditions built into Definition 2.1 below, and the characterization [1, Theorem 7.4 (kernel cocycle)] that the integrated kernel deficit (17) vanishes if and only if the family is divisible, proved there under a uniform bound (C1) and operator-norm continuity (C2)—both of which every family in this paper satisfies by definition, the uniform bound with constant $\sqrt{N}$ by finite-dimensionality (see below) and the continuity by Definition 2.1. No argument below invokes either result as a proof step; both enter only through the stated definitions and the interpretation of the integrated deficit as an indivisibility functional.

2.1 Two-time families and the indivisibility deficit

Let $\Omega = \{1, \dots, N\}$ be a finite configuration space, μ counting measure on Ω, and $\mathcal{H} = L^2(\Omega, \mu) \cong \mathbb{C}^N$ with the standard inner product. We write the action of a column-stochastic matrix $\Gamma \in \mathbb{R}^{N\times N}_{\geq 0}$ on $\mathcal{H}$ via $(\Gamma f)(i) = \sum_j \Gamma_{ij} f(j)$, so that the column-sum normalization $\sum_i \Gamma_{ij} = 1$ corresponds to $\Gamma^* \mathbf{1} = \mathbf{1}$ where $\mathbf{1} = (1, \dots, 1)$. Let $[0, T]^2_{\leq} = \{(t, s) : 0 \leq s \leq t \leq T\}$.

Definition 2.1 (Two-time column-stochastic family)**.** A *two-time column-stochastic family* on Ω over horizon $T > 0$ is a map $\Gamma\colon [0, T]^2_{\leq} \to \mathbb{R}^{N\times N}_{\geq 0}$ such that for all $(t, s) \in [0, T]^2_{\leq}$, $\Gamma(t \leftarrow s)$ is column-stochastic, $\Gamma(s \leftarrow s) = I$, and the map $(t, s) \mapsto \Gamma(t \leftarrow s)$ is continuous in operator norm.

The boundary and continuity conditions of Definition 2.1 are the conditions (R1)–(R2) of [1, Theorem 5.4]; under them the family is in particular Borel-measurable and uniformly bounded in operator norm by $\sqrt{N}$, since each column of a column-stochastic matrix has Euclidean norm at most 1, so $\|\Gamma\|_{\mathrm{op}} \leq \|\Gamma\|_F \leq \sqrt{N}$.

Definition 2.2 (Deficit operator, deficit, and deficit at resolution)**.** For a two-time family Γ and a triple (t, s, r) with $0 \leq s < r < t \leq T$, the *deficit operator* at (t, s, r) is

$$\Delta(t, s, r; \Gamma) \;=\; \Gamma(t \leftarrow s) - \Gamma(t \leftarrow r)\,\Gamma(r \leftarrow s) \;\in\; \mathcal{B}(\mathcal{H}), \tag{2}$$

and the *indivisibility deficit* at (t, s, r) is its operator norm,

$$D(t, s, r; \Gamma) \;=\; \|\Delta(t, s, r; \Gamma)\|_{\mathrm{op}}\,. \tag{3}$$

At temporal resolution $\delta t > 0$, the *deficit at resolution* δt is

$$D_{\mathrm{CK}}(\delta t; \Gamma) \;=\; \sup_{\substack{(t,s,r):\\ s<r<t,\ t-s\le\delta t}} D(t,s,r;\Gamma). \tag{4}$$

The deficit at resolution δt is the worst-case operator-norm failure of CK factorization across triples of duration at most δt. The family is divisible if and only if $D_{\mathrm{CK}}(\delta t; \Gamma) = 0$ for all $\delta t \in (0, T]$.

Lemma 2.3 (Attainment of the resolution deficit)**.** *For a two-time family* Γ *(Definition 2.1) and any* $\delta t \in (0, T]$*, the map* $(t, s, r) \mapsto \Delta(t, s, r; \Gamma)$ *extends operator-norm continuously to the compact set*

$$K(\delta t) \;=\; \{(t,s,r) : 0 \le s \le r \le t \le T,\ t - s \le \delta t\},$$

and the extension vanishes on the degenerate boundary faces $r = s$*,* $r = t$*, and* $t = s$*. Consequently the supremum in (4) over the set of strict triples equals the maximum of the continuous extension over* $K(\delta t)$*; in particular it is finite and attained.*

Proof. The formula (2) is defined and operator-norm continuous on all of $K(\delta t)$, since Γ is norm-continuous and uniformly bounded and products of uniformly bounded norm-continuous families are norm-continuous. On the boundary faces the boundary condition $\Gamma(u \leftarrow u) = I$ gives: at $r = s$, $\Delta = \Gamma(t \leftarrow s) - \Gamma(t \leftarrow s)\, I = 0$; at $r = t$, $\Delta = \Gamma(t \leftarrow s) - I\, \Gamma(t \leftarrow s) = 0$; and $t = s$ forces $r = s = t$, where $\Delta = I - I = 0$. The function $D = \|\Delta\|_{\mathrm{op}}$ is continuous and nonnegative on the compact set $K(\delta t)$ and vanishes on the three degenerate faces. If its maximum over $K(\delta t)$ is positive, every maximizer lies off the faces and is therefore a strict triple, so the supremum over the strict triples equals the maximum; if the maximum is zero, then $D \equiv 0$ and both quantities vanish. In either case the supremum in (4) equals the maximum over $K(\delta t)$, finite and attained. □

Remark 2.4 (Hierarchy with non-Markovianity). By [1, Proposition 2.6], the Markov property implies CK factorization, but two-time CK factorization does not imply Markovianity: the canonical witness is Feller's construction [6] of a non-Markovian process with the semigroup property, whose two-point transition probabilities compose exactly while the three-time joint distributions are inconsistent with any Markov chain sharing those transitions. (Projections of Markov chains under lumpable partitions do not witness this converse: a lumpable partition yields an observed process that is itself Markov.) Indivisibility—failure of CK at the two-time level—is therefore strictly stronger than non-Markovianity [1, Corollary 2.7]. The perturbation theorems of this paper are statements about D_{CK} and apply to the strictly indivisible class.

2.2 The divisible reference family

We model an interaction with the process that is conditioned on the instantaneous state at the observation time *and* dynamically consistent across time. State conditioning alone does not produce divisibility: a collection of kernels, each conditioned on the present state, need not compose coherently along intermediate times, and a family assembled from such kernels can violate CK. Dynamical consistency—the requirement that the interaction's action factor through every choice of intermediate time—is exactly the divisibility property, and we impose it as the defining structural hypothesis.

Definition 2.5 (Divisible reference family)**.** A *divisible reference family* on Ω over horizon T is a two-time column-stochastic family Γ_{ref} that is divisible: $\Gamma_{\mathrm{ref}}(t \leftarrow s) = \Gamma_{\mathrm{ref}}(t \leftarrow r)\, \Gamma_{\mathrm{ref}}(r \leftarrow s)$ for all $0 \le s \le r \le t \le T$.

A wide class of dynamically consistent, state-conditioned mechanisms produces such a family: any time-inhomogeneous Markov process on Ω with norm-continuous transition kernel induces, via $\Gamma_{\mathrm{ref};ij}(t \leftarrow s) = \mathbb{P}(Y_t = i \mid Y_s = j)$, a divisible two-time family in the sense of Definition 2.5. The structural content of the choice is not the specific reference kernel but the divisibility itself, which is what the perturbation theory uses; a state-conditioned interaction that fails dynamical consistency falls outside the model.

2.3 The convex perturbation

Definition 2.6 (Convex perturbation)**.** Let Γ_0 be a two-time column-stochastic family and Γ_{ref} a divisible reference family on the same Ω and horizon T. For each $g \in [0,1]$, the *convex perturbation at coupling* g is the two-time family

$$\Gamma_g(t \leftarrow s) \;=\; (1-g)\,\Gamma_0(t \leftarrow s) \;+\; g\,\Gamma_{\mathrm{ref}}(t \leftarrow s), \qquad (t,s) \in [0,T]^2_{\leq}. \tag{5}$$

The convex combination is column-stochastic for each $g \in [0,1]$, and the family Γ_g inherits the boundary and continuity conditions of Definition 2.1 from Γ_0 and Γ_{ref}. The parameter g measures the fraction of the perturbed family contributed by the divisible reference; we call it the *observer coupling* as a label only.

Remark 2.7 (Structural reading of the perturbation). The convex form is the canonical model. Any dynamically consistent state-conditioned interaction that contributes a fixed fraction g of the transition mass and leaves the remaining $1-g$ to the indivisible substrate produces, at the level of two-time kernels, the family of Definition 2.6. State-dependent or time-dependent coupling functions $g(t,s,x)$ generalize the construction; the uniform-coupling convex family is the leading case and the one for which the perturbation theory is cleanest. The state-dependent case is treated, now with a rigorous bound, in Section 6.

Remark 2.8 (The perturbation is not a divisibility extension). The convex perturbation does not augment the configuration space or pass to a divisible Markov representation on a larger state space, in the sense of the projection-operator and hidden-Markov constructions [7, 8, 9]. Such augmentations produce a divisible kernel on a larger $\widetilde{\Omega} \supsetneq \Omega$ whose marginal on Ω is the original indivisible family; the augmentation eliminates the deficit by construction. The convex perturbation operates on Ω itself and adds a divisible component in convex measure: the perturbed family Γ_g remains a column-stochastic family on the original Ω, and its deficit is the object of study. The two constructions are conceptually orthogonal.

3 The deficit identity and the suppression bound

The first principal result is an exact algebraic identity for the deficit operator of the perturbed family, together with the suppression bound it implies in operator norm. The identity is the organizing result of the paper; the suppression bound is its triangle-inequality consequence, the nonmonotonicity of the response (Remark 3.2) is its exact algebraic consequence, and the two-threshold theory of Section 4 is the quantitative form both of them force.

Theorem 3.1 (Deficit identity and suppression bound)**.** *Let Γ_0 be a two-time column-stochastic family on Ω over horizon T and let Γ_{ref} be a divisible reference family on the same (Ω, T). For each $g \in [0,1]$, let Γ_g be the convex perturbation of Definition 2.6. Then for all triples (t,s,r) with $0 \leq s < r < t \leq T$, the deficit operator at coupling g admits the algebraic identity*

$$\Delta(t,s,r;\Gamma_g) \;=\; (1-g)\,\big(A - CE\big) \;-\; g(1-g)\,(C-D)(F-E), \tag{6}$$

where $A = \Gamma_0(t \leftarrow s)$, $C = \Gamma_0(t \leftarrow r)$, $D = \Gamma_{\mathrm{ref}}(t \leftarrow r)$, $E = \Gamma_0(r \leftarrow s)$, *and* $F = \Gamma_{\mathrm{ref}}(r \leftarrow s)$. *Consequently,*

$$D(t,s,r;\Gamma_g) \;\le\; (1-g)\,D(t,s,r;\Gamma_0) \;+\; g(1-g)\,M(t,s,r), \tag{7}$$

where

$$M(t,s,r) \;=\; \|(C-D)(F-E)\|_{\mathrm{op}} \;=\; \big\|\big(\Gamma_0(t \leftarrow r) - \Gamma_{\mathrm{ref}}(t \leftarrow r)\big)\big(\Gamma_{\mathrm{ref}}(r \leftarrow s) - \Gamma_0(r \leftarrow s)\big)\big\|_{\mathrm{op}}. \tag{8}$$

In particular, $g \mapsto \Delta(t,s,r;\Gamma_g)$ *is a polynomial in* g, $D(t,s,r;\Gamma_g)$ *is continuous in* g *on* $[0,1]$, *and* $D(t,s,r;\Gamma_1) = 0$.

Proof. Fix (t,s,r) and adopt the abbreviations of the statement, together with $B = \Gamma_{\mathrm{ref}}(t \leftarrow s)$. Divisibility of Γ_{ref} gives $B = DF$. Direct expansion of the convex combination yields

$$\begin{aligned}
&\Gamma_g(t \leftarrow s) - \Gamma_g(t \leftarrow r)\,\Gamma_g(r \leftarrow s) \\
&\qquad = \big[(1-g)A + gB\big] - \big[(1-g)C + gD\big]\big[(1-g)E + gF\big] \\
&\qquad = (1-g)A + g\,DF - (1-g)^2 CE - g(1-g)(CF + DE) - g^2 DF.
\end{aligned}$$

Collecting the DF terms gives $g\,DF - g^2 DF = g(1-g)\,DF$, and splitting $(1-g)A = (1-g)^2 A + g(1-g)A$:

$$\Delta(t,s,r;\Gamma_g) \;=\; (1-g)^2(A - CE) \;+\; g(1-g)\big[A + DF - CF - DE\big].$$

The bracketed term equals $(A - CE) + (CE + DF - CF - DE) = (A - CE) - (C-D)(F-E)$, since $(C-D)(F-E) = CF - CE - DF + DE$. Substituting,

$$\Delta(t,s,r;\Gamma_g) \;=\; \big[(1-g)^2 + g(1-g)\big](A - CE) \;-\; g(1-g)(C-D)(F-E),$$

and $(1-g)^2 + g(1-g) = (1-g)\big[(1-g)+g\big] = (1-g)$, which gives (6). The triangle inequality applied to (6) yields (7), since $\|A - CE\|_{\mathrm{op}} = D(t,s,r;\Gamma_0)$ and $\|(C-D)(F-E)\|_{\mathrm{op}} = M(t,s,r)$ by definition. The polynomial dependence on g is manifest in (6); continuity of $D(t,s,r;\Gamma_g)$ in g follows, and at $g = 1$ the right-hand side of (6) vanishes identically. □

Remark 3.2 (The interaction term, and nonmonotonicity of the response). The quantity $M(t,s,r)$ defined in (8) factors as the operator norm of a product of two differences: the substrate–reference difference on the late interval and the substrate–reference difference on the early interval (with sign reversed). Submultiplicativity of the operator norm gives the explicit upper bound

$$M(t,s,r) \;\le\; \|\Gamma_0(t \leftarrow r) - \Gamma_{\mathrm{ref}}(t \leftarrow r)\|_{\mathrm{op}} \cdot \|\Gamma_0(r \leftarrow s) - \Gamma_{\mathrm{ref}}(r \leftarrow s)\|_{\mathrm{op}}.$$

In particular, when the substrate and reference are uniformly close in operator norm, M is small as a product of two small factors: the interaction term enters the suppression bound at second order in the substrate–reference distance, whereas the substrate deficit enters at first order.

The suppression bound (7), viewed as a function of $g \in [0,1]$, is in general *not* monotonically nonincreasing. Differentiation of $(1-g)\,D(t,s,r;\Gamma_0) + g(1-g)\,M(t,s,r)$ gives derivative $-D(t,s,r;\Gamma_0) + (1-2g)\,M(t,s,r)$, which is non-positive on all of $[0,1]$ if and only if $D(t,s,r;\Gamma_0) \ge M(t,s,r)$. When the interaction term dominates the substrate deficit at the triple, the upper bound first increases on a neighborhood of $g = 0$ before being driven to zero by the $(1-g)$ factor as $g \to 1$. The exact deficit need not respond monotonically either: in the interaction-dominant regime of the example of Section 4.3 it vanishes in the interior of the coupling range and rebounds before its final descent. This regime is not pathological: it occurs whenever the substrate and reference are far enough apart that the late- and early-interval differences combine constructively in the operator-norm product. The substantive content of the

suppression bound is therefore not monotonicity in g but vanishing at $g = 1$ and the explicit functional form on $[0, 1]$. This observation is load-bearing: it is what makes the first-crossing threshold operationally insufficient and forces the terminal threshold of Section 4.

Corollary 3.3 (Suppression at resolution)**.** *Under the hypotheses of Theorem 3.1, for each $\delta t \in (0, T]$ and each $g \in [0, 1]$,*

$$D_{\mathrm{CK}}(\delta t; \Gamma_g) \;\le\; (1-g)\, D_{\mathrm{CK}}(\delta t; \Gamma_0) \;+\; g(1-g)\, M_*(\delta t), \tag{9}$$

where $M_*(\delta t) = \sup_{\{(t,s,r):\, s<r<t,\ t-s\le\delta t\}} M(t, s, r)$, finite and attained by the argument of Lemma 2.3 applied to the norm-continuous map $(t, s, r) \mapsto (C - D)(F - E)$.

Proof. Take the supremum over the triple set in the bound of Theorem 3.1; the coefficients $(1 - g)$ and $g(1 - g)$ are uniform in the triple. □

4 Detection thresholds: terminal suppression and interval-local stability

The suppression bound of Theorem 3.1 is a triangle-inequality estimate, and by Remark 3.2 the response of the deficit to the coupling need not be monotone. A single "critical coupling" defined as a first crossing therefore does not support the operational reading one wants from a threshold—a first crossing can, in general, be followed by re-entry above the detection level. The detectability theory accordingly runs on two thresholds.

4.1 The two thresholds and the terminal suppression theorem

Definition 4.1 (First-crossing and terminal suppression thresholds)**.** Let Γ_0, Γ_{ref}, and Γ_g be as in Theorem 3.1, fix a resolution $\delta t \in (0, T]$ with $D_{\mathrm{CK}}(\delta t; \Gamma_0) > 0$, and fix a detection threshold $\varepsilon \in (0, D_{\mathrm{CK}}(\delta t; \Gamma_0))$. Define

$$g_{\mathrm{first}}(\delta t; \varepsilon) \;=\; \inf\big\{g \in [0, 1] : D_{\mathrm{CK}}(\delta t; \Gamma_g) < \varepsilon\big\}, \tag{10}$$

$$g_{\mathrm{term}}(\delta t; \varepsilon) \;=\; \inf\big\{g \in [0, 1] : D_{\mathrm{CK}}(\delta t; \Gamma_h) < \varepsilon \;\text{ for all } h \in [g, 1]\big\}. \tag{11}$$

The first-crossing threshold records where the deficit first dips below the detection level; the terminal threshold records where it goes below the detection level *and stays there*. Only the latter supports the operational reading "above this coupling, indivisibility is undetectable at sensitivity ε." Both sets in Definition 4.1 are nonempty—the coarse bound in the proof below shows the deficit is eventually below ε—so both thresholds are well defined, and trivially $g_{\mathrm{first}} \le g_{\mathrm{term}}$, since every g in the terminal set lies in the first-crossing set.

Lemma 4.2 (Lipschitz dependence on the coupling)**.** *For fixed δt, the map $g \mapsto D_{\mathrm{CK}}(\delta t; \Gamma_g)$ is Lipschitz on $[0, 1]$ with constant $D_{\mathrm{CK}}(\delta t; \Gamma_0) + M_*(\delta t)$.*

Proof. From the identity (6), for $g, g' \in [0, 1]$ and any triple in the resolution set,

$$\Delta(t, s, r; \Gamma_g) - \Delta(t, s, r; \Gamma_{g'}) \;=\; (g' - g)(A - CE) \;-\; (g - g')\big(1 - (g + g')\big)(C - D)(F - E),$$

using $g(1-g) - g'(1-g') = (g-g')(1-(g+g'))$. Since $|1-(g+g')| \le 1$ on $[0, 1]^2$, taking norms gives $\big|D(t, s, r; \Gamma_g) - D(t, s, r; \Gamma_{g'})\big| \le |g-g'|\,\big[D(t, s, r; \Gamma_0) + M(t, s, r)\big] \le |g-g'|\,\big[D_{\mathrm{CK}}(\delta t; \Gamma_0) + M_*(\delta t)\big]$ uniformly over the triple set, and the same bound passes to the supremum. □

Theorem 4.3 (Terminal suppression threshold: existence and universal bound)**.** *Under the hypotheses of Definition 4.1, set*

$$g_0 \;:=\; 1 \;-\; \frac{\varepsilon}{D_{\mathrm{CK}}(\delta t;\Gamma_0)+M_*(\delta t)} \;\in\; (0,1).$$

Then:

(i) (Decreasing envelope) For every $g\in[0,1]$,

$$D_{\mathrm{CK}}(\delta t;\Gamma_g) \;\le\; (1-g)\big[D_{\mathrm{CK}}(\delta t;\Gamma_0)+g\,M_*(\delta t)\big] \;\le\; B(g) := (1-g)\big[D_{\mathrm{CK}}(\delta t;\Gamma_0)+M_*(\delta t)\big],$$

and the coarse envelope B is strictly decreasing on $[0,1]$ with $B(g_0)=\varepsilon$.

(ii) (Terminal suppression) $D_{\mathrm{CK}}(\delta t;\Gamma_g)\le\varepsilon$ for every $g\in[g_0,1]$, and $D_{\mathrm{CK}}(\delta t;\Gamma_g)<\varepsilon$ strictly for every $g\in(g_0,1]$.

(iii) (Threshold chain and universal bound)

$$0 \;<\; g_{\mathrm{first}}(\delta t;\varepsilon) \;\le\; g_{\mathrm{term}}(\delta t;\varepsilon) \;\le\; g_0 \;=\; 1-\frac{\varepsilon}{D_{\mathrm{CK}}(\delta t;\Gamma_0)+M_*(\delta t)} \;<\; 1.$$

Proof. (i) Setting $u=1-g\in[0,1]$ in Corollary 3.3, the right-hand side of (9) is $u\,D_{\mathrm{CK}}(\delta t;\Gamma_0)+(1-u)u\,M_*(\delta t) = u\big[D_{\mathrm{CK}}(\delta t;\Gamma_0)+(1-u)M_*(\delta t)\big] \le u\big[D_{\mathrm{CK}}(\delta t;\Gamma_0)+M_*(\delta t)\big] = B(g)$, using $1-u\le 1$. The envelope $B(g)$ is linear in g with strictly negative slope $-[D_{\mathrm{CK}}(\delta t;\Gamma_0)+M_*(\delta t)]<0$, hence strictly decreasing, and $B(g_0)=(1-g_0)[D_{\mathrm{CK}}+M_*]=\varepsilon$ by the definition of g_0.

(ii) For $g\ge g_0$, monotonicity of B gives $D_{\mathrm{CK}}(\delta t;\Gamma_g)\le B(g)\le B(g_0)=\varepsilon$; for $g>g_0$ strictly, $B(g)<B(g_0)=\varepsilon$, so the deficit is strictly below the detection threshold. (The strictness step matters: the bound proves only $\le\varepsilon$ at g_0 itself, and the strict inequality defining the threshold sets is obtained on the half-open interval $(g_0,1]$, whose infimum still recovers g_0 in (iii).)

(iii) By (ii), every $g\in(g_0,1]$ belongs to the terminal set of (11): for all $h\in[g,1]\subset(g_0,1]$, $D_{\mathrm{CK}}(\delta t;\Gamma_h)<\varepsilon$. Hence the terminal set contains $(g_0,1]$ and its infimum is at most g_0, which gives $g_{\mathrm{term}}\le g_0<1$; in particular both sets in Definition 4.1 are nonempty, so the thresholds are well defined, and $g_{\mathrm{first}}\le g_{\mathrm{term}}$ as noted. Finally $g_{\mathrm{first}}>0$: at $g=0$, $D_{\mathrm{CK}}(\delta t;\Gamma_0)>\varepsilon$ by hypothesis, and Lemma 4.2 gives $D_{\mathrm{CK}}(\delta t;\Gamma_g)\ge D_{\mathrm{CK}}(\delta t;\Gamma_0)-g\,[D_{\mathrm{CK}}(\delta t;\Gamma_0)+M_*(\delta t)]>\varepsilon$ for every $g<\eta := (D_{\mathrm{CK}}(\delta t;\Gamma_0)-\varepsilon)/(D_{\mathrm{CK}}(\delta t;\Gamma_0)+M_*(\delta t))$, so the first-crossing set is contained in $[\eta,1]$ and $g_{\mathrm{first}}\ge\eta>0$. □

Remark 4.4 (Operational reading; the diagnostic role of the first crossing). Below g_{first}, the deficit at resolution δt is at least ε and indivisibility remains detectable at that sensitivity. Above g_{term}, the deficit is permanently below ε and the perturbed family is operationally indistinguishable from a divisible family at that sensitivity, for every stronger coupling as well. Between the two thresholds the response may be nonmonotone, and in general a first crossing can be followed by re-entry above the detection level; every operational statement of the form "above this coupling indivisibility is undetectable" is therefore a statement about g_{term}, never about g_{first}. The first-crossing threshold retains a diagnostic role—it is the quantity read off directly from a plotted response curve, and the worked examples of Section 4.3 report it—but it is not interpreted terminally anywhere in this paper. In both regimes of the two-state example of Section 4.3 the response happens not to re-enter, so the two thresholds coincide there; the general warning stands.

Remark 4.5 (The bound is sign-blind in the interaction term). The bound of Theorem 4.3(iii) is the leading-order estimate from the convex-combination structure and gives a *sufficient* coupling for permanent suppression below ε. Its dependence on the interaction scale M_* should be read carefully: as M_* increases, $1-\varepsilon/(D_{\mathrm{CK}}+M_*)$ increases toward 1, so a larger interaction term

makes the sufficient bound *weaker*—the guarantee is pushed to stronger coupling—and does not show suppression at smaller coupling. The exact interaction term in the identity (6) may align destructively with the substrate deficit and lower the *actual* thresholds substantially (the interaction-dominant regime of the example in Section 4.3 has $g_{\mathrm{term}} \approx 0.108$ against the bound 0.834); but the norm bound discards sign and cannot prove that. When $M_*(\delta t)$ is small, the bound reduces to $g_{\mathrm{term}} \leq 1 - \varepsilon/D_{\mathrm{CK}}(\delta t; \Gamma_0)$, the pure linear suppression rate dictated by the convex combination.

Remark 4.6 (Mixing properties of the reference family). The interaction scale $M_*(\delta t)$ admits explicit upper bounds when the reference family is the semigroup of a uniformly ergodic Markov chain. If $\Gamma_{\mathrm{ref}}(t \leftarrow s) = e^{(t-s)Q_{\mathrm{ref}}}$ for an irreducible generator Q_{ref} with stationary distribution π_{ref} (written as a column vector), the standard mixing-time estimates [34, 35, 36, 37] give

$$\left\| \Gamma_{\mathrm{ref}}(t \leftarrow s) - \pi_{\mathrm{ref}} \mathbf{1}^\top \right\|_{\mathrm{op}} \leq C(\pi_{\mathrm{ref}})\, e^{-\lambda(t-s)}, \qquad t - s \geq 0,$$

where $\lambda > 0$ is the spectral gap of Q_{ref} and $C(\pi_{\mathrm{ref}}) \geq 1$ is a constant depending on the chain (the Fill bound [36] gives $C = \sqrt{\pi_{\mathrm{ref,min}}^{-1} - 1}$ in the $\ell^2(\pi_{\mathrm{ref}})$ norm; $\ell^2 \to$ op conversion adds at most a $\sqrt{\pi_{\mathrm{ref,min}}^{-1}}$ factor). Note the orientation: for a *column*-stochastic ergodic family every column converges to π_{ref}, so the long-time limit is the matrix $\pi_{\mathrm{ref}}\mathbf{1}^\top$ whose columns all equal π_{ref}. Combined with submultiplicativity (Remark 3.2), this gives quantitative control of $M_*(\delta t)$ that is dimension-explicit and scales correctly under refinement of the resolution. We do not use this bound in the threshold theorem; it is recorded for the reader who wishes to estimate the thresholds in concrete examples.

4.2 Interval-local stability of detectability

The terminal theorem bounds the thresholds from above. The complementary question is a lower bound: can an arbitrarily weak coupling destroy detectability? On any fixed compact interval of resolutions on which the unperturbed deficit stays uniformly above the detection level, the answer is no, with an explicit constant. We state the result in the honest form it deserves: it is a stability statement obtained from uniform separation and continuity on a compact set—not a resolution–coupling scaling law, and not a dynamical speed limit. On any compact rectangle of parameters bounded away from the degenerate ends, a positive lower bound of this kind follows from the uniform threshold gap; the content is the stability of detectability under weak coupling, together with the explicit constant.

Lemma 4.7 (Joint continuity of the resolution deficit)**.** *The map* $(\delta t, g) \mapsto D_{\mathrm{CK}}(\delta t; \Gamma_g)$ *is jointly continuous on* $(0, T] \times [0, 1]$*, and* $\delta t \mapsto M_*(\delta t)$ *is continuous on* $(0, T]$*.*

Proof. The map $(t, s, r, g) \mapsto D(t, s, r; \Gamma_g)$ is jointly continuous on $K(T) \times [0, 1]$: by (6) it is, for fixed (t, s, r), the norm of a polynomial in g whose coefficient operators depend norm-continuously on (t, s, r) (Lemma 2.3), with all quantities uniformly bounded. The correspondence $\delta t \mapsto K(\delta t)$ of Lemma 2.3 is compact-valued and continuous in the Hausdorff sense on $(0, T]$: it is nested increasing in δt, and any triple with $t - s \leq \delta t$ is approximated, for $\delta t' < \delta t$ close, by the affine contraction of the triple toward its own midpoint, which changes it by $O(\delta t - \delta t')$; upper hemicontinuity is immediate from closedness of the constraint $t - s \leq \delta t$. Berge's maximum theorem [38, Ch. VI] (see also [39, Theorem 17.31]) then yields joint continuity of the value function $(\delta t, g) \mapsto \max_{K(\delta t)} D(\cdot; \Gamma_g) = D_{\mathrm{CK}}(\delta t; \Gamma_g)$, the equality of the maximum over $K(\delta t)$ with the supremum over the strict triples being Lemma 2.3. The same argument applied to the norm-continuous map $(t, s, r) \mapsto (C - D)(F - E)$ gives continuity of M_*. □

Assumption 4.8 (Uniform detectability on a resolution interval)**.** Fix a nondegenerate interval $[\delta t_-, \delta t_+] \subset (0, T]$ and a detection threshold $\varepsilon > 0$ such that

$$m_0 \;:=\; \min_{\delta t \in [\delta t_-, \delta t_+]} D_{\mathrm{CK}}(\delta t; \Gamma_0) \;>\; \varepsilon,$$

the minimum existing by Lemma 4.7. A convenient sufficient condition is a two-sided scaling regime: if there are exponents $\alpha, \beta > 0$ and constants $c_\alpha, C_\alpha, c_\beta, C_\beta > 0$ with

$$c_\alpha\,(\delta t)^\alpha \;\le\; D_{\mathrm{CK}}(\delta t; \Gamma_0) \;\le\; C_\alpha\,(\delta t)^\alpha, \tag{12}$$

$$c_\beta\,(\delta t)^\beta \;\le\; M_*(\delta t) \;\le\; C_\beta\,(\delta t)^\beta \tag{13}$$

on $[\delta t_-, \delta t_+]$, then every $\varepsilon < c_\alpha(\delta t_-)^\alpha$ satisfies the gap condition. Under the hypotheses of the local scaling law of [1, Proposition 4.12, with the indivisibility tensor of Definition 4.11], the substrate exponent α is governed by the order of vanishing of the indivisibility tensor at the relevant time, and $\beta = \alpha$ generically for Lipschitz families. We emphasize the demoted role of the exponents: they serve to guarantee the uniform detectability gap $m_0 > \varepsilon$ and, in Proposition 4.9, to furnish envelope constants for the interval-dependent upper bound; they generate no scaling law, and the stability lower bound does not use them.

Proposition 4.9 (Interval-local stability of the detection threshold)**.** *Under Assumption 4.8, let* $\overline{M} := \max_{[\delta t_-, \delta t_+]} M_*(\delta t)$ *(finite by Lemma 4.7). Define*

$$\eta(\varepsilon) \;:=\; \begin{cases} \dfrac{m_0 + \overline{M} - \sqrt{(m_0 - \overline{M})^2 + 4\,\overline{M}\varepsilon}}{2\,\overline{M}}, & \overline{M} > 0, \\[2ex] 1 - \dfrac{\varepsilon}{m_0}, & \overline{M} = 0. \end{cases} \tag{14}$$

Then $\eta(\varepsilon) \in (0, 1)$ *and, for every* $\delta t \in [\delta t_-, \delta t_+]$,

$$g_{\mathrm{first}}(\delta t; \varepsilon) \;\ge\; \eta(\varepsilon), \qquad \textit{hence} \qquad g_{\mathrm{first}}(\delta t; \varepsilon) \cdot \delta t \;\ge\; K_-(\varepsilon) := \eta(\varepsilon)\,\delta t_- \;>\; 0. \tag{15}$$

The complementary upper bound is interval-uniform but interval-dependent: from Theorem 4.3(iii),

$$g_{\mathrm{first}}(\delta t; \varepsilon) \cdot \delta t \;\le\; g_{\mathrm{term}}(\delta t; \varepsilon) \cdot \delta t \;\le\; \delta t \left(1 - \tfrac{\varepsilon}{D_{\mathrm{CK}}(\delta t; \Gamma_0) + M_*(\delta t)}\right), \tag{16}$$

which under the scaling regime (12)–(13) is at most $\delta t_+ - \varepsilon\,\delta t_- \big[C_\alpha(\delta t_+)^\alpha + C_\beta(\delta t_+)^\beta\big]^{-1}$*. The upper bound depends on the endpoints of the chosen interval; on a fixed horizon it is trivially bounded by* T*, and no unbounded growth occurs unless the horizon itself is sent to infinity, a regime not treated here.*

Proof. Fix $\delta t \in [\delta t_-, \delta t_+]$ and let (t^*, s^*, r^*) be a strict triple attaining $D_{\mathrm{CK}}(\delta t; \Gamma_0)$, which exists by Lemma 2.3 (the maximum of the continuous extension lies off the degenerate faces because it is positive). The reverse triangle inequality applied to the identity (6) at this triple gives, for every $g \in [0, 1]$,

$$D_{\mathrm{CK}}(\delta t; \Gamma_g) \;\ge\; D(t^*, s^*, r^*; \Gamma_g) \;\ge\; (1-g)\,D_{\mathrm{CK}}(\delta t; \Gamma_0) - g(1-g)\,M(t^*, s^*, r^*) \;\ge\; (1-g)\big[m_0 - g\,\overline{M}\big],$$

using $1 - g \ge 0$ with $D_{\mathrm{CK}}(\delta t; \Gamma_0) \ge m_0$ in the first term, and $g(1-g) \ge 0$ with $M(t^*, s^*, r^*) \le M_*(\delta t) \le \overline{M}$ in the second. The right-hand side exceeds ε exactly when the quadratic

$$q(g) \;:=\; \overline{M} g^2 - (m_0 + \overline{M})\,g + (m_0 - \varepsilon)$$

is strictly positive. For $\overline{M} > 0$, q is an upward parabola with $q(0) = m_0 - \varepsilon > 0$ and $q(1) = -\varepsilon < 0$, so its smaller root is the $\eta(\varepsilon)$ of (14) (the discriminant simplifies as $(m_0 + \overline{M})^2 - 4\overline{M}(m_0 - \varepsilon) =$

$(m_0 - \overline{M})^2 + 4\overline{M}\varepsilon)$, it lies in $(0,1)$, and on $[0,1]$ one has $q > 0$ precisely on $[0, \eta(\varepsilon))$. For $\overline{M} = 0$ the condition is linear: $(1-g)m_0 > \varepsilon$ iff $g < 1 - \varepsilon/m_0 = \eta(\varepsilon)$. In either case, $D_{\mathrm{CK}}(\delta t; \Gamma_g) > \varepsilon$ for every $g \in [0, \eta(\varepsilon))$ and every δt in the interval, so the first-crossing set of (10) is contained in $[\eta(\varepsilon), 1]$ and $g_{\mathrm{first}}(\delta t; \varepsilon) \geq \eta(\varepsilon)$. Multiplying by $\delta t \geq \delta t_-$ gives (15).

For the upper bound, multiply the chain of Theorem 4.3(iii) by δt; under the scaling regime, bound $D_{\mathrm{CK}} + M_* \leq C_\alpha(\delta t)^\alpha + C_\beta(\delta t)^\beta$, use $\delta t \leq \delta t_+$ on the leading term and $\delta t \geq \delta t_-$ with monotonicity of the denominator on the correction term. □

Remark 4.10 (What the stability result does and does not say). The proposition says: on a resolution band where the unperturbed deficit is uniformly detectable, no coupling weaker than the explicit $\eta(\varepsilon)$ can push the deficit below the detection level anywhere on the band. It does not say that a resolution–coupling trade-off with universal constant governs the dynamics: the constant $K_-(\varepsilon)$ is manufactured from the interval endpoints and the substrate–reference pair, the coupling g is a mixture parameter rather than a dynamical rate, and no $\delta t \to 0$ asymptotics is analyzed. The product form invites a comparison with speed-limit inequalities; that comparison, and its limits, are confined to Section 7.3.

Remark 4.11 (A dimensionally explicit rate scale). The product $g_{\mathrm{first}} \cdot \delta t$ is dimensionally a (dimensionless coupling) × (time). To express the trade-off in rate form, define the local rate operators of the two families as the one-sided diagonal derivatives

$$L_0(t) \;:=\; \partial_u \, \Gamma_0(u \leftarrow t)\big|_{u=t^+}, \qquad L_{\mathrm{ref}}(t) \;:=\; \partial_u \, \Gamma_{\mathrm{ref}}(u \leftarrow t)\big|_{u=t^+},$$

whenever these limits exist in operator norm. For the divisible reference, L_{ref} is its generator family in the ordinary sense; for the indivisible substrate no generator reconstructing the two-time family exists [1], and $L_0(t)$ is only the local rate of the family at the diagonal—the word "generator" is reserved for the reference. Assume both diagonal derivatives exist, with

$$0 \;<\; \Lambda \;:=\; \sup_{t \in [0,T]} \|L_0(t) - L_{\mathrm{ref}}(t)\|_{\mathrm{op}} \;<\; \infty,$$

the strict positivity being a nondegeneracy requirement (a reference whose local rates coincide everywhere with the substrate's carries no leading-order contrast). The perturbation rate scale $\gamma_g := g\,\Lambda$ has dimensions of inverse time, and Proposition 4.9 transfers to the dimensionless statement $\gamma_{g_{\mathrm{first}}(\delta t;\varepsilon)} \cdot \delta t \geq \Lambda\, K_-(\varepsilon)$ on the resolution interval. The constant remains non-universal; see Section 7.3.

4.3 An explicit two-state illustration

To make the deficit identity, the suppression bound, and the two-threshold theory concrete, we work out an example on the configuration space $\Omega = \{1, 2\}$ at the time triple $(s, r, t) = (0, 1, 2)$. The example exhibits both the substrate-dominant and the interaction-dominant regimes, including a nonmonotone response. It is specified at this single triple, so the threshold notions are read pointwise throughout the subsection: $D(t, s, r; \Gamma_g)$ and $M(t, s, r)$ stand in for $D_{\mathrm{CK}}(\delta t; \cdot)$ and $M_*(\delta t)$, and every statement of Sections 3 and 4 holds verbatim under this substitution, a single triple being the one-point case of the supremum over triples.

Substrate. Define the substrate Γ_0 at the three time points by

$$E := \Gamma_0(r \leftarrow s) = \begin{pmatrix} 0.8 & 0.2 \\ 0.2 & 0.8 \end{pmatrix}, \qquad C := \Gamma_0(t \leftarrow r) = \begin{pmatrix} 0.7 & 0.3 \\ 0.3 & 0.7 \end{pmatrix}, \qquad A := \Gamma_0(t \leftarrow s) = \begin{pmatrix} 0.58 & 0.42 \\ 0.42 & 0.58 \end{pmatrix}.$$

Direct computation yields $CE = \left(\begin{smallmatrix} 0.62 & 0.38 \\ 0.38 & 0.62 \end{smallmatrix}\right)$, which differs from A, so Γ_0 is indivisible at (s, r, t) with substrate deficit

$$D(t, s, r; \Gamma_0) \;=\; \|A - CE\|_{\mathrm{op}} \;=\; \left\| 0.04 \cdot \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix} \right\|_{\mathrm{op}} \;=\; 0.08.$$

Reference. Take Γ_{ref} to be the symmetric two-state Glauber semigroup with rate $\lambda > 0$,

$$\Gamma_{\mathrm{ref}}(\tau) \;=\; \tfrac{1}{2} \begin{pmatrix} 1 + e^{-2\lambda\tau} & 1 - e^{-2\lambda\tau} \\ 1 - e^{-2\lambda\tau} & 1 + e^{-2\lambda\tau} \end{pmatrix},$$

which is divisible with stationary distribution $\pi_{\mathrm{ref}} = (1/2, 1/2)$. For the slow-mixing case $\lambda = 0.5$, $D := \Gamma_{\mathrm{ref}}(1) = F \approx \left(\begin{smallmatrix} 0.684 & 0.316 \\ 0.316 & 0.684 \end{smallmatrix}\right)$, and

$$(C - D)(F - E) \;=\; -\,0.01606 \cdot 0.11606 \cdot 2 \cdot \begin{pmatrix} 1 & -1 \\ -1 & 1 \end{pmatrix} \;\approx\; 0.00373 \cdot \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix},$$

so the interaction magnitude is $M(t, s, r) \approx 0.00746$.

Verification of the deficit identity. Both $A - CE$ and $(C - D)(F - E)$ are scalar multiples of the same rank-one matrix $\left(\begin{smallmatrix} -1 & 1 \\ 1 & -1 \end{smallmatrix}\right)$. The deficit operator at coupling g predicted by (6) is therefore

$$\Delta(t, s, r; \Gamma_g) \;=\; \big[(1 - g)(0.04) - g(1 - g)(0.00373)\big] \cdot \begin{pmatrix} -1 & 1 \\ 1 & -1 \end{pmatrix}.$$

At $g = 0.5$ the bracket equals 0.0191, giving a predicted deficit operator $\left(\begin{smallmatrix} -0.0191 & 0.0191 \\ 0.0191 & -0.0191 \end{smallmatrix}\right)$. Direct evaluation gives $\Gamma_{0.5}(t \leftarrow s) \approx \left(\begin{smallmatrix} 0.574 & 0.426 \\ 0.426 & 0.574 \end{smallmatrix}\right)$ and $\Gamma_{0.5}(t \leftarrow r)\,\Gamma_{0.5}(r \leftarrow s) \approx \left(\begin{smallmatrix} 0.593 & 0.407 \\ 0.407 & 0.593 \end{smallmatrix}\right)$, with difference $\approx \left(\begin{smallmatrix} -0.019 & 0.019 \\ 0.019 & -0.019 \end{smallmatrix}\right)$. The two computations agree to display precision.

Substrate-dominant regime. Since $0.00373 < 0.04$, the deficit at coupling g in operator norm is

$$D(t, s, r; \Gamma_g) \;=\; 2(1 - g)\,(0.04 - 0.00373\,g),$$

which is strictly decreasing on $[0, 1]$ (consistent with Remark 3.2, since $D(t, s, r; \Gamma_0) = 0.08 > M(t, s, r) \approx 0.0075$ places this triple firmly in the substrate-dominant regime). A strictly decreasing response crosses the detection level once and never returns, so here the two thresholds of Definition 4.1 coincide, $g_{\mathrm{first}} = g_{\mathrm{term}}$. For detection threshold $\varepsilon = 0.04$, the quadratic $0.00373\,g^2 - 0.04373\,g + 0.02 = 0$ gives the common threshold $g_{\mathrm{first}} = g_{\mathrm{term}} \approx 0.477$, against the terminal-threshold bound of Theorem 4.3(iii) of $1 - \varepsilon/[D(t, s, r; \Gamma_0) + M(t, s, r)] \approx 0.543$. The bound is sufficient and tight to about 14% in this regime.

Interaction-dominant regime. With the slow-mixing reference replaced by a fast-mixing one, $\lambda = 2$ (giving $\Gamma_{\mathrm{ref}}(1) \approx \left(\begin{smallmatrix} 0.509 & 0.491 \\ 0.491 & 0.509 \end{smallmatrix}\right)$), and the substrate unchanged, $M(t, s, r)$ rises to approximately $0.222 > D(t, s, r; \Gamma_0) = 0.08$. In this regime the upper bound of Theorem 3.1 is nonmonotone in g near $g = 0$ (Remark 3.2); the exact deficit, given by $2(1 - g)\,|0.04 - 0.111\,g|$, is instead nonmonotone in the interior: it falls to a first crossing of $\varepsilon = 0.05$ at $g_{\mathrm{first}} \approx 0.108$, vanishes at $g \approx 0.36$, then traces a *nonmonotone rebound* peaking near $g \approx 0.68$ at deficit value ≈ 0.023, and descends to zero at $g = 1$. The rebound peak 0.023 stays below the selected threshold 0.05, so no re-entry above the detection level occurs and the terminal threshold coincides with the first crossing here, $g_{\mathrm{term}} = g_{\mathrm{first}} \approx 0.108$. (At a more sensitive threshold $\varepsilon < 0.023$ the rebound *would* re-cross, the two thresholds would separate, and only g_{term} would certify permanent suppression—which is the general warning of Remark 4.4 in concrete form.)

The universal bound of Theorem 4.3(iii), $1 - 0.05/0.302 \approx 0.834$, remains valid but is loose here: the interaction term aligns destructively with the substrate deficit and drives the actual thresholds far below the sign-blind norm bound, as Remark 4.5 anticipates. This variant also illustrates the regime in which the cone-geometric characterization of divisibility for two-state stochastic-quantum dynamics in [40] provides an alternative description of when the interaction term dominates; the formal connection between the cone characterization and the operator-norm thresholds here is an open question.

5 Causal organization on $L^2([0,T]\times\Omega)$

The perturbation theorems of Sections 3–4 are statements about pointwise triples (t,s,r) of the two-time family. They lift to one-sided causal operators on $L^2([0,T]\times\Omega)$ via the retarded transition operator of [1, Section 7].

5.1 The perturbed retarded transition operator

Let $\mathcal{H}_T = L^2([0,T]\times\Omega)$ be the Hilbert space of square-integrable spacetime observables. Let $\mathcal{R}_0$ be the retarded transition operator of Γ_0 in the sense of [1, Definition 7.1], with kernel $K_{\mathcal{R}_0}(t,x;s,y) = \mathbf{1}_{\{s\le t\}}\Gamma_{0,xy}(t \leftarrow s)$, and similarly $\mathcal{R}_{\mathrm{ref}}$ for Γ_{ref}; both are bounded on $\mathcal{H}_T$ with norm at most $\sqrt{N}\,T$ by the uniform bound of Section 2 and [1, Lemma 7.2].

Definition 5.1 (Perturbed retarded transition operator)**.** For $g\in[0,1]$, the *perturbed retarded transition operator* at coupling g is the bounded linear operator $\mathcal{R}_g\colon \mathcal{H}_T\to\mathcal{H}_T$ with kernel

$$K_{\mathcal{R}_g}(t,x;s,y) \;=\; \mathbf{1}_{\{s\le t\}}\,\Gamma_{g,xy}(t\leftarrow s) \;=\; (1-g)K_{\mathcal{R}_0}(t,x;s,y) + gK_{\mathcal{R}_{\mathrm{ref}}}(t,x;s,y).$$

Proposition 5.2 (Causality of $\mathcal{R}_g$)**.** *For each $g\in[0,1]$, the operator $\mathcal{R}_g$ is causal with respect to the filtration $\{\mathcal{F}_\tau = L^2([0,\tau]\times\Omega)\}_{\tau\in[0,T]}$: for all $\tau\in[0,T]$ and $f\in\mathcal{H}_T$,*

$$\mathbf{1}_{[0,\tau]}\cdot\mathcal{R}_g f \;=\; \mathbf{1}_{[0,\tau]}\cdot\mathcal{R}_g\big(\mathbf{1}_{[0,\tau]}\cdot f\big).$$

Equivalently, the future subspaces $\{f\in\mathcal{H}_T : f=0$ a.e. on $[0,\tau]\times\Omega\}$, $\tau\in[0,T]$, are invariant under $\mathcal{R}_g$.

Proof. Direct from the kernel structure: $K_{\mathcal{R}_g}(t,x;s,y)=0$ for $s>t$, so $(\mathcal{R}_g f)(t,x)$ depends only on $f(s,y)$ for $s\le t$. Multiplication by $\mathbf{1}_{[0,\tau]}(t)$ on either side restricts the range and preserves the dependence structure. If moreover f vanishes a.e. on $[0,\tau]\times\Omega$, the identity gives $\mathbf{1}_{[0,\tau]}\cdot\mathcal{R}_g f = 0$, which is the invariance of the future subspaces. □

5.2 The integrated kernel deficit under perturbation

Following [1, Section 7], the *integrated kernel deficit* of a two-time family Γ is

$$J(\Gamma) \;=\; \int_0^T\!\!\int_0^T\!\!\int_0^T \mathbf{1}_{\{s<r<t\}}\,D(t,s,r;\Gamma)^2\,ds\,dr\,dt. \tag{17}$$

By [1, Theorem 7.4 (kernel cocycle)], $J(\Gamma)=0$ if and only if Γ is divisible; the hypotheses of that theorem—the uniform bound (C1) and operator-norm continuity (C2)—hold for every family in this paper by Definition 2.1 and the $\sqrt{N}$ bound of Section 2.

Theorem 5.3 (Quadratic suppression of the integrated deficit)**.** *Under the hypotheses of Theorem 3.1, for each $g \in [0,1]$,*

$$J(\Gamma_g) \;\le\; 2(1-g)^2\, J(\Gamma_0) \;+\; 2g^2(1-g)^2\, J_M, \tag{18}$$

where $J_M = \int_0^T \int_0^T \int_0^T \mathbf{1}_{\{s<r<t\}}\, M(t,s,r)^2\, ds\, dr\, dt$ is finite under the hypotheses of Definition 2.1. In particular $J(\Gamma_g) = O\big((1-g)^2\big)$ as $g \to 1$: the upper bound vanishes at rate $(1-g)^2$.

Proof. By Theorem 3.1, $D(t,s,r;\Gamma_g) \le (1-g)D(t,s,r;\Gamma_0) + g(1-g)M(t,s,r)$. Squaring and using $(a+b)^2 \le 2a^2 + 2b^2$,

$$D(t,s,r;\Gamma_g)^2 \;\le\; 2(1-g)^2 D(t,s,r;\Gamma_0)^2 + 2g^2(1-g)^2 M(t,s,r)^2.$$

Integrating over the simplex $\{(t,s,r) : 0 \le s < r < t \le T\}$ and using the definitions of $J(\Gamma_g), J(\Gamma_0)$ from (17) and of J_M from the statement yields (18). Finiteness of J_M follows from $M(t,s,r) \le \|C-D\|_{\mathrm{op}} \cdot \|F-E\|_{\mathrm{op}} \le 4N$ uniformly, via the Frobenius bound $\|\cdot\|_{\mathrm{op}} \le 2\sqrt{N}$ on differences of column-stochastic matrices; with $\int_0^T \int_0^T \int_0^T \mathbf{1}_{\{s<r<t\}}\, ds\, dr\, dt = T^3/6$, this gives $J_M \le 16N^2T^3/6 < \infty$. Both terms of (18) carry the factor $(1-g)^2$, so the upper bound vanishes at rate $(1-g)^2$ as $g \to 1$; this is an upper rate, since cancellation between the two terms of the identity (6) can make $J(\Gamma_g)$ itself decay faster (Remark 3.2), and the theorem asserts $J(\Gamma_g) = O((1-g)^2)$, not an exact rate. □

The integrated deficit $J(\Gamma_g)$ is a single scalar functional of the family that quantifies indivisibility on the trajectory space $\mathcal{H}_T$ in the sense of [1, Theorem 7.4]. Theorem 5.3 establishes that this trajectory-space scalar is suppressed at least quadratically in $(1-g)$ as $g \to 1$, providing the trajectory-space shadow of the pointwise threshold theory.

6 Robustness: state-dependent and nonlinear coupling

The convex perturbation of Definition 2.6 is the leading-order model of a dynamically consistent state-conditioned interaction. This section treats two natural extensions. The state-dependent extension, previously available only as a heuristic, is now a proposition with explicit constants: the column-wise structure of the coupling makes the finite-dimensional bound elementary once the perturbation is written as a diagonal deformation of a constant-coupling family.

6.1 State-dependent coupling: a rigorous deficit bound

Let $g\colon [0,T]^2_{\le} \times \Omega \to [0,1]$ be any function, with

$$g_{\min} := \inf_{(t,s,x)} g(t,s,x), \qquad g_{\max} := \sup_{(t,s,x)} g(t,s,x), \qquad \delta_g := g_{\max} - g_{\min}.$$

The state-dependent perturbation couples column j at strength $g(t,s,j)$:

$$\Gamma_{g(\cdot)}(t \leftarrow s)_{ij} \;=\; \big(1 - g(t,s,j)\big)\, \Gamma_{0,ij}(t \leftarrow s) \;+\; g(t,s,j)\, \Gamma_{\mathrm{ref};ij}(t \leftarrow s). \tag{19}$$

Each column of $\Gamma_{g(\cdot)}(t \leftarrow s)$ is a convex combination of the corresponding columns of Γ_0 and Γ_{ref}, so $\Gamma_{g(\cdot)}$ is column-stochastic, and $\Gamma_{g(\cdot)}(s \leftarrow s) = I$ since both constituent families satisfy the boundary condition. No regularity of the coupling function is needed for the pointwise bound below; if g is continuous the perturbed family is norm-continuous and the resolution-level supremum is attained as in Lemma 2.3.

Proposition 6.1 (State-dependent coupling: rigorous deficit bound)**.** *With* $G(t,s) := \mathrm{diag}\big(g(t,s,1),\dots,g(t,s,$ *the state-dependent perturbation admits the factorization*

$$\Gamma_{g(\cdot)}(t \leftarrow s) \;=\; \Gamma_0(t \leftarrow s)\,\big(I - G(t,s)\big) \;+\; \Gamma_{\mathrm{ref}}(t \leftarrow s)\,G(t,s), \tag{20}$$

and for every triple $0 \le s < r < t \le T$*,*

$$D(t,s,r;\Gamma_{g(\cdot)}) \;\le\; (1-g_{\min})\,D(t,s,r;\Gamma_0) \;+\; g_{\max}(1-g_{\min})\,M(t,s,r) \;+\; R(\delta_g), \tag{21}$$

with the explicit remainder

$$R(\delta_g) \;=\; \frac{\delta_g}{2}\,\big(1{+}2\sqrt{N}\big)\,W_* + \Big(\frac{\delta_g\,W_*}{2}\Big)^2, \qquad W_* \;:=\; \sup_{(t,s)\in[0,T]^2_{\le}} \|\Gamma_{\mathrm{ref}}(t \leftarrow s) - \Gamma_0(t \leftarrow s)\|_{\mathrm{op}} \;\le\; 2\sqrt{N}. \tag{22}$$

In particular $R(\delta_g) = O(\delta_g)$ *as* $\delta_g \to 0$*, uniformly over triples, and the bound* (21) *passes to the resolution level by taking suprema. For constant coupling* ($\delta_g = 0$) *the bound reduces to Theorem 3.1.*

Proof. The factorization (20) is entrywise identical to (19): right multiplication by the diagonal $G(t,s)$ rescales column j by $g(t,s,j)$.

Fix the constant reference coupling $\bar g := (g_{\min} + g_{\max})/2 \in [0,1]$ and let $\Gamma_{\bar g}$ be the constant-coupling perturbation of Definition 2.6. From (20),

$$P(t,s) \;:=\; \Gamma_{g(\cdot)}(t \leftarrow s) - \Gamma_{\bar g}(t \leftarrow s) \;=\; \big(\Gamma_{\mathrm{ref}}(t \leftarrow s) - \Gamma_0(t \leftarrow s)\big)\,\big(G(t,s) - \bar g I\big),$$

and since the operator norm of a diagonal matrix is its largest absolute entry,

$$\|P(t,s)\|_{\mathrm{op}} \;\le\; W_* \cdot \max_j \big|g(t,s,j) - \bar g\big| \;\le\; W_*\,\frac{\delta_g}{2} \;=:\; \rho,$$

the choice of $\bar g$ as the midpoint halving the deviation. Abbreviate $P_{ts} = P(t,s)$, $P_{tr} = P(t,r)$, $P_{rs} = P(r,s)$. Expanding the composition,

$$\Delta(t,s,r;\Gamma_{g(\cdot)}) \;=\; \Delta(t,s,r;\Gamma_{\bar g}) \;+\; P_{ts} \;-\; P_{tr}\,\Gamma_{\bar g}(r \leftarrow s) \;-\; \Gamma_{\bar g}(t \leftarrow r)\,P_{rs} \;-\; P_{tr}P_{rs}.$$

Every constant-coupling matrix is column-stochastic, hence of operator norm at most $\sqrt{N}$, so taking norms,

$$D(t,s,r;\Gamma_{g(\cdot)}) \;\le\; D(t,s,r;\Gamma_{\bar g}) \;+\; \rho \;+\; 2\sqrt{N}\,\rho \;+\; \rho^2 \;=\; D(t,s,r;\Gamma_{\bar g}) + R(\delta_g).$$

Theorem 3.1 at the constant coupling $\bar g$ gives $D(t,s,r;\Gamma_{\bar g}) \le (1-\bar g)\,D(t,s,r;\Gamma_0) + \bar g(1-\bar g)\,M(t,s,r)$, and the monotone replacements $1-\bar g \le 1-g_{\min}$ and $\bar g(1-\bar g) \le g_{\max}(1-g_{\min})$ (both factors nonnegative, $\bar g \le g_{\max}$, $1-\bar g \le 1-g_{\min}$) yield (21). The bound on W_* is the Frobenius estimate on differences of column-stochastic matrices used in Theorem 5.3. □

Remark 6.2. Proposition 6.1 settles affirmatively the question left open in the previous version of this paper, where the estimate (21) was displayed as a heuristic with an unproved order claim for the remainder: the column-wise structure of the coupling does support a rigorous finite-dimensional bound, with the explicit constants of (22). Consequently the terminal-threshold machinery of Section 4 applies to the state-dependent family through the envelope $(1-g_{\min})D_{\mathrm{CK}}(\delta t;\Gamma_0) + g_{\max}(1-g_{\min})M_*(\delta t) + R(\delta_g)$: driving $g_{\min} \to 1$ forces the deficit below any threshold. A genuinely state-dependent threshold theory—couplings tailored per column to minimize the deficit at fixed budget—is a separate optimization question we do not pursue.

6.2 Nonlinear coupling

A nonlinear coupling of strength parameter g is a perturbation $\Gamma_{f(g)}$ with $f\colon [0,1] \to [0,1]$ continuous and nondecreasing, $f(0) = 0$, $f(1) = 1$. The deficit responds at strength $f(g)$: $D(t,s,r;\Gamma_{f(g)}) \le (1 - f(g))\, D(t,s,r;\Gamma_0) + f(g)(1 - f(g))\, M(t,s,r)$. Because a continuous monotone f need not be injective, thresholds transform through the generalized inverse

$$f^{\leftarrow}(y) \;:=\; \inf\{g \in [0,1] : f(g) \ge y\},$$

rather than through f^{-1}. With $g_0 = 1 - \varepsilon/[D_{\mathrm{CK}}(\delta t;\Gamma_0) + M_*(\delta t)]$ the universal level of Theorem 4.3: for every $\gamma \in (g_0, 1]$ and every $g \ge f^{\leftarrow}(\gamma)$, monotonicity and continuity of f give $f(g) \ge \gamma > g_0$, hence $D_{\mathrm{CK}}(\delta t;\Gamma_{f(g)}) < \varepsilon$ strictly by Theorem 4.3(ii); therefore the terminal threshold in the g-parametrization satisfies $g^{(f)}_{\mathrm{term}}(\delta t;\varepsilon) \le f^{\leftarrow}(\gamma)$ for every $\gamma \in (g_0, 1]$, and when f is strictly increasing this collapses to $g^{(f)}_{\mathrm{term}} \le f^{-1}(g_0)$. The qualitative structure of the two-threshold theory is thus invariant under monotone reparametrization of the coupling, which is the relevant invariance class for the structural conclusions of the paper.

7 Discussion

7.1 Relation to divisibility-based non-Markovianity functionals

Remark 7.1 (Relation to non-Markovianity functionals). The divisibility-based non-Markovianity functionals of Wolf–Eisert–Cubitt–Cirac [10], Rivas–Huelga–Plenio [12, 13], Chruściński–Maniscalco [14], and Buscemi–Datta [15] are typically defined on quantum dynamical maps; the joint quantum/classical formulation of Vacchini–Smirne–Laine–Piilo–Breuer [16] and the classical/quantum P-divisibility transfer of Benatti–Chruściński–Nichele [17] provide the bridge to the column-stochastic setting of the present paper. The vanishing of the deficit at $g = 1$ implies, for any non-Markovianity functional continuous with respect to the operator-norm topology on two-time families used here, that the functional is correspondingly suppressed as $g \to 1$; for the integrated deficit J of [1, Section 7], Theorem 5.3 provides the quantitative form. We do not claim monotonicity of such functionals in g—the nonmonotonicity of Remark 3.2 argues against expecting it—and the coupling at which a given functional becomes undetectable is a terminal-threshold question in the sense of Definition 4.1, dependent on the functional's sensitivity to operator-norm perturbations. By the Buscemi–Datta [15] equivalence between divisibility and monotonic information decrease, exact monotonicity of every data-processing-compatible information measure is restored only where the deficit vanishes identically, that is, at $g = 1$; above g_{term} the deficit is suppressed below ε but need not vanish, so the terminal threshold marks ε-level suppression of the deficit rather than exact restoration of information monotonicity. A quantitative transfer from the deficit level to bounds on the monotonicity violations of specific information measures is not pursued here.

7.2 Distinction from the process-tensor framework

The non-Markovianity literature using the process-tensor framework [23, 5, 24, 25] treats memory effects in CPTP dynamics through a multi-time Stinespring/dilation construction operating on discrete sequences of control operations. The present paper differs structurally on the four counts set out in the introduction. The substantive implication is that the convex-perturbation theorem of Section 3 does not have a direct process-tensor analogue: process-tensor non-Markovianity is quantified by deviation from a Markov-order property, not by an operator-norm CK deficit on a continuous two-time family. The settings are complementary rather than competing; the column-stochastic setting here is upstream of the CPTP setting,

in that any column-stochastic family is a diagonal-CPTP specialization in the sense of the diagonal-retention lift [1, Definition 5.1], and the threshold theory specializes in that direction.

7.3 Product-form bounds and speed-limit theory

Proposition 4.9 bounds the product $g_{\text{first}} \cdot \delta t$ below on a compact resolution interval. Product-form inequalities between a cost and a temporal increment are the signature of dynamical speed-limit theory: the Mandelstam–Tamm [26] and Margolus–Levitin [27] bounds in the quantum setting, their classical-stochastic analogues by Shiraishi–Funo–Saito [28], Shanahan–Chenu–Margolus–del Campo [29], and Wu–Yu [30], the Fisher-information unification of Nishiyama–Hasegawa [33], and the driven-system and review literature [31, 32]. We record the formal resemblance and disclaim any deeper identification. The constant $K_-(\varepsilon)$ arises from uniform separation and continuity on a chosen compact interval and is not universal; the coupling g is a mixture parameter, not a dynamical speed, energy dispersion, statistical distance, activity, or entropy-production rate; no $\delta t \to 0$ asymptotics is analyzed; and no geometric path length is bounded. The result is an interval-local stability statement for the detectability of indivisibility, not a dynamical speed limit, and we do not assert a common underlying geometry. Whether a genuine rate-form or information-geometric formulation exists—a derivation of a stability constant from a Fisher-information cost on the perturbation, or conversely an operator-norm perturbation derivation of a speed-limit bound—is left as future work and is not claimed here.

7.4 Relation to convex resource theory of non-Markovianity

The convex resource theory of Bhattacharya–Bhattacharya–Majumdar [22] establishes free operations (Markovian semigroups) and a robustness measure of non-Markovianity defined via convex combinations of dynamical maps, with monotonicity under free operations. The setting differs from the present one in that [22] works on bounded time intervals with quantum dynamical maps, whereas the present paper works on a two-time column-stochastic family with a continuous resolution parameter δt. The convex perturbation of Definition 2.6 is the column-stochastic analogue of the resource-theoretic convex combination of [22], and the terminal-threshold theorem is in the same family as the robustness-of-non-Markovianity result there. The principal substantive distinction is that the present thresholds are on the deficit at fixed resolution δt, not on a resource-theoretic functional defined on a fixed time interval. The two results are mutually compatible and structurally parallel; we anticipate that a formal connection through the diagonal-CPTP specialization of [1, Definition 5.1] can be established, but defer this to companion work.

7.5 The thresholds and the structural interpretation of indivisibility

The two-threshold theory provides a quantitative form of the structural claim that the empirical detectability of indivisibility depends on the perturbation budget available to the observer. Below $g_{\text{first}}(\delta t; \varepsilon)$, the deficit at resolution δt is at least the detection threshold ε and the family is operationally indivisible. Above $g_{\text{term}}(\delta t; \varepsilon)$, the deficit remains below ε at every stronger coupling, and the family is operationally indistinguishable from a divisible family at the prescribed sensitivity—permanently, which is what the operational reading requires (Remark 4.4). The transition is structural rather than merely empirical: the perturbation modifies the family, not the apparatus.

This structural reading clarifies a recurring question in the framework of [1] and [2]: when is indivisibility a property of the dynamics versus a property of the observation? The threshold theory answers: it is a property of the (substrate, perturbation, resolution, sensitivity) quadru-

ple. The substrate's deficit and the perturbation's coupling jointly determine, at given resolution and sensitivity, whether indivisibility is detected. The operator-theoretic content—the concentration of the deficit on a closed input/output subspace pair and the obstruction to divisible approximation, in the sense of [1, Theorem 4.5 (deficit concentration and obstruction to divisible approximation)]—is independent of the perturbation; the empirical detectability of that content depends on the perturbation budget. The geometric characterization of divisibility for two-state stochastic-quantum dynamics due to Pimenta [40] is structurally compatible with the present operator-norm formulation; the formal connection between the Pimenta cone-geometric characterization and the present threshold theory at $N = 2$ is an open question.

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